

# COMP3011 Coursework 1 — API Documentation

Travel Planner API (FastAPI + SQLAlchemy + SQLite)  
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This document describes a subset of key REST endpoints for the Travel Planner API. The interactive Swagger UI is available at **/docs** when the server is running.

**Base URL (local):** `http://127.0.0.1:8000`

**Content-Type:** `application/json`

**Authentication:** none (coursework prototype). A lightweight `user_name` field is used to simulate user interactions.

## Common status codes:

- **200** OK (successful read / analytics)
- **201** Created (successful creation)
- **204** No Content (successful deletion)
- **400** Bad Request (e.g., invalid trip date range)
- **404** Not Found (resource does not exist)
- **409** Conflict (duplicate bookmark)
- **422** Validation Error (input schema violation)

## Key Endpoints

### ***POST /places — Create Place***

**Purpose:** Create a new place (point of interest).

**Request example:**

```
{"city": "Paris", "name": "Louvre Museum", "category": "museum"}
```

**Response example:**

```
{"id": 1, "city": "Paris", "name": "Louvre Museum", "category": "museum"}
```

**Status codes:** 201 Created, 422 Validation Error

### ***GET /places?city=Paris&category=museum — List Places***

**Purpose:** List places, optionally filtered by city and/or category.

**Request:** (none)

**Response example:**

```
[{"id": 1, "city": "Paris", "name": "Louvre Museum", "category": "museum"}]
```

**Status codes:** 200 OK

### ***POST /trips — Create Trip***

**Purpose:** Create a trip with an inclusive date range.

**Request example:**

```
{"name": "Paris Weekend", "start_date": "2026-03-01", "end_date": "2026-03-03"}
```

**Response example:**

```
{"id": 1, "name": "Paris Weekend", "start_date": "2026-03-01", "end_date": "2026-03-03"}
```

**Status codes:** 201 Created, 400 Bad Request, 422 Validation Error

### ***POST /trips/{trip\_id}/places — Add Place to Trip***

**Purpose:** Add a place to a trip. This creates an itinerary item (TripPlace).

**Request example:**

```
{"place_id": 1, "day": 1, "planned_order": 1, "note": "Arrive early"}
```

**Response example:**

```
{"id": 1, "trip_id": 1, "place_id": 1, "day": 1, "planned_order": 1, "note": "Arrive early"}
```

**Status codes:** 201 Created, 404 Not Found, 422 Validation Error

### ***GET /trips/{trip\_id}/places — List Trip Itinerary***

**Purpose:** List itinerary items for a trip (sorted by day and planned\_order).

**Request:** (none)

**Response example:**

```
[{"id": 1, "trip_id": 1, "place_id": 1, "day": 1, "planned_order": 1, "note": "Arrive early"}]
```

**Status codes:** 200 OK, 404 Not Found

### ***POST /places/{place\_id}/bookmark — Create Bookmark***

**Purpose:** Bookmark a place for a given user\_name (unique per user\_name + place\_id).

**Request example:**

```
{"user_name": "Han"}
```

**Response example:**

```
{"id": 1, "place_id": 1, "user_name": "Han", "created_at": "2026-03-01T10:00:00"}
```

**Status codes:** 201 Created, 404 Not Found, 409 Conflict, 422 Validation Error

### ***POST /places/{place\_id}/reviews — Create Review***

**Purpose:** Create a review (rating 1-5) for a place.

**Request example:**

```
{"user_name": "Han", "rating": 5, "comment": "Excellent."}
```

**Response example:**

```
{"id": 1, "place_id": 1, "user_name": "Han", "rating": 5, "comment": "Excellent.", "created_at": "2026-03-01T10:10:00"}
```

**Status codes:** 201 Created, 404 Not Found, 422 Validation Error

### ***POST /trips/{trip\_id}/expenses — Create Expense***

**Purpose:** Record an expense for a trip (used for budget analytics).

**Request example:**

```
{"category": "food", "amount": 18.0, "currency": "EUR", "date": "2026-03-01", "note": "Dinner"}
```

**Response example:**

```
{"id": 1, "trip_id": 1, "category": "food", "amount": 18.0, "currency": "EUR", "date": "2026-03-01", "note": "Dinner"}
```

**Status codes:** 201 Created, 404 Not Found, 422 Validation Error

### ***GET /analytics/trips/{trip\_id}/budget-summary?currency=EUR — Budget Summary***

**Purpose:** Return total spending and totals grouped by category for a trip.

**Request:** (none)

**Response example:**

```
{"trip_id": 1, "currency": "EUR", "total": 42.8, "by_category": [{"category": "ticket", "total": 22.0}, {"category": "food", "total": 18.0}, {"category": "transport", "total": 2.8}]}
```

**Status codes:** 200 OK, 404 Not Found

### ***GET /analytics/cities/{city}/top-bookmarked?limit=10 — Top Bookmarked (City)***

**Purpose:** Rank places within a city by bookmark count (descending).

**Request:** (none)

**Response example:**

```
[{"place_id": 1, "city": "Paris", "name": "Louvre Museum", "category": "museum", "value": 3.0, "count": null}]
```

**Status codes:** 200 OK

### ***GET /analytics/cities/{city}/top-rated?limit=10&min;\_reviews=3 — Top Rated (City)***

**Purpose:** Rank places within a city by average rating, requiring a minimum review count.

**Request:** (none)

**Response example:**

```
[{"place_id": 1, "city": "Paris", "name": "Louvre Museum", "category": "museum", "value": 4.67, "count": 3}]
```

**Status codes:** 200 OK

### ***POST /analytics/trips/{trip\_id}/generate-itinerary — Generate Itinerary***

**Purpose:** Assign day values to itinerary items with day = null using a simple round-robin strategy.

**Request example:**

(no body)

**Response example:**

```
{"trip_id":1,"updated_items":2}
```

**Status codes:** 200 OK, 404 Not Found, 400 Bad Request